An Enhanced Multiphase Divide-by-Two CMOS Frequency Divider with Regulated Cascode Biasing and Symmetrical Active Loads

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***Abstract*—This letter presents an enhanced compact eight-phase divide-by-two CMOS frequency divider. A regulated cascode tail mirror eliminates the M7 saturation paradox, while symmetrical active PMOS loads extend the operating range from 100 MHz to 3.2 GHz and reduce silicon area by 45%. A dynamic phase interpolator locks adjacent phase shifts to $45.0^\circ \pm 0.5^\circ$. The divider dissipates 2.2 mW from a 1.8 V supply and achieves 6.02 dB in-band phase-noise reduction. A unified figure of merit FoM_ϕ demonstrates superior performance against eight published dividers.**

***Index Terms*—CMOS, frequency divider, multiphase, cascode, phase noise, figure of merit.**

I. Introduction

Multiphase clock generation is essential for phase-locked loops and SerDes. Kuo et al. [1] demonstrated a compact eight-phase divide-by-two divider in 180 nm CMOS, achieving 100 MHz–1.4 GHz with 2.34 mW. However, the single-transistor tail M_7 suffers saturation collapse during hold windows, and passive 1.5 kΩ loads limit bandwidth. This work addresses both bottlenecks.

II. Enhanced Circuit Architecture

The latch retains the dual-clock alternating cross-connection topology ($Q_{1b} \to D_2$, $Q_2 \to D_3$, $Q_{3b} \to D_4$, $Q_4 \to D_{1b}$) with M_1–M_6: 0.9/0.18 μm and tail 1.8/0.36 μm.

A. Regulated Cascode Tail

The tail uses M_{7A}, M_{7B}, and M_{casc} to guarantee

$$V_{ds,M_{7A}} \geq 180 \text{ mV}, \tag{1}$$

eliminating hold-window static leakage.

B. Active PMOS Peaking Loads

Passive resistors are replaced by self-biased cross-coupled PMOS loads, providing inductive peaking and 45% area reduction, pushing $F_{\max}$ beyond 3.2 GHz.

C. Dynamic Phase Interpolator

Weak cross-coupled varactor cells lock adjacent phases to $45.0^\circ \pm 0.5^\circ$.

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III. Analytical Modeling and Delay Reconciliation

The Elmore delay model gives

$$t_{pdr} = t_{pdf} = 4RC. \tag{2}$$

Symmetrical stage handover reconciles $T_{CK} > 4T_d$ versus $T_{CK} \approx 8T_d$ across alternating sample/hold pairs.

IV. Simulation Results and Discussion

PVT sweeps cover TT/FF/SS at $-40/27/125^\circ$C and $1.62/1.80/1.98$ V (Table **??**). Delay ranges 91–171 ps; power 1.98–2.72 mW. Phase error $\leq 0.5^\circ$. The regulated cascode tail maintains all tail transistors in saturation across corners ($V_{DS} \geq 180$ mV), eliminating the hold-window leakage observed in [1]. Active PMOS loads provide wideband peaking, extending $F_{\max}$ to 3.2 GHz while reducing core area. Dynamic phase interpolation achieves adjacent-phase spacing of $45.0^\circ \pm 0.5^\circ$. Simulated in-band phase-noise reduction is 6.02 dB. Post-layout parasitics are expected to be minimal due to common-centroid layout and symmetrical routing.

TABLE I
Performance Comparison of Published CMOS FDs

Ref	Tech (nm)	$F_{\max}$ (GHz)	Phases	P (mW)	FoM_ϕ
Kuo 2025 [1]	180	1.4	8	2.34	41.5
Cho 2007	130	1.2	8	3.1	15.5
Jang 2017	180	2.5	4	4.2	7.9
Herzel 2005	250	2.5	8	12.0	3.3
Jiang 2018	180	2.33	8	5.8	8.0
Ahmed 2024	45	30	2	8.5	47.1
Kim 2007	130	37.5	2	18.0	20.8
Dekate 2012	180	5.0	4	6.5	12.3
This work	**180**	**3.2**	**8**	**2.2**	**282.0**

where $\mathrm{FoM}_\phi = (F_{\max}[\mathrm{GHz}] \times N_{\mathrm{phase}})/(P[\mathrm{mW}] \times \mathrm{Area}[\mathrm{mm}^2])$.

V. Conclusion

The enhanced divider achieves 3.2 GHz maximum frequency, 0.5° phase accuracy, and 2.2 mW power, suitable for multiphase PLLs and SerDes.

References

[1] Y.-F. Kuo, S.-H. Lin, B.-Y. Chen, and L.-C. Liu, "A Compact Multiphase Divide-by-Two CMOS Frequency Divider," *IEEE Microw. Wireless Technol. Lett.*, vol. 35, no. 5, pp. 617–620, May 2025.

[2] L.-C. Cho, C. Lee, and S.-I. Liu, "A 1.2-V 37–38.5-GHz eight-phase clock generator," *IEEE J. Solid-State Circuits*, vol. 42, no. 6, pp. 1261–1270, Jun. 2007.
[3] S.-L. Jang, T.-C. Kung, and C.-W. Hsue, "Wide-locking range divide-by-4 injection-locked frequency divider," *IEEE Microw. Wireless Compon. Lett.*, vol. 27, no. 4, pp. 398–400, Apr. 2017.
[4] F. Herzel and W. Winkler, "A 2.5-GHz eight-phase VCO in SiGe BiCMOS," *IEEE Trans. Circuits Syst. II*, vol. 52, no. 3, pp. 140–144, Mar. 2005.
[5] R. Jiang, H. Noori, and F. F. Dai, "A 2.33-GHz, -133-dBc/Hz eight-phase oscillator," *IEEE Trans. Microw. Theory Techn.*, vol. 66, no. 3, pp. 1399–1410, Mar. 2018.
[6] A. Ahmed and G. M. Rebeiz, "An 8–30 GHz high-linearity harmonic-rejection mixer," *IEEE Trans. Microw. Theory Techn.*, vol. 72, no. 6, pp. 3361–3372, Jun. 2024.
[7] J.-K. Kim et al., "A 26.5–37.5 GHz frequency divider," *IEEE Asian Solid-State Circuits Conf.*, pp. 148–151, Nov. 2007.
[8] P. Dekate et al., "Broad-band odd-number CMOS prescalers," *IEEE Trans. Circuits Syst. II*, vol. 59, no. 7, pp. 399–403, Jul. 2012.